

Network Security: Detecting ARP Spoofing

Objective

The objective of this lab was to detect ARP spoofing, identify the attacker, and understand its impact on network traffic, including potential denial-of-service (DoS) conditions.

Steps

1. Opening Wireshark

Launched Wireshark and opened the provided pcap file containing the network capture data.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	PCSSystemtec_53:0c:...	Broadcast	ARP	60	Who has 192.168.10.4? Tell 192.168.10.5
2	1.006646	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
3	3.006692	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
4	5.010103	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
5	7.018900	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
6	9.038853	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
7	11.091956	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
8	13.091549	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
9	15.101815	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
10	17.107401	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
11	19.108114	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
12	21.110135	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
13	23.118700	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
14	25.126325	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
15	27.126418	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
16	29.137086	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
17	31.142099	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
18	33.141829	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
19	35.153856	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
20	37.163628	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba
21	39.169156	PCSSystemtec_53:0c:...	Netgear_e2:d5:c3	ARP	60	192.168.10.4 is at 08:00:27:53:0c:ba

2. Detecting Duplicate ARP Values

Analyzed the ARP packets in the pcap file to find duplicate ARP replies indicating the presence of spoofing. ARP spoofing often results in multiple devices claiming the same IP address with different MAC addresses.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	PCSSystemtec_53:0c:...	Broadcast	ARP	60	Who has 192.168.10.4? Tell 192.168.10.5
26	48.614791	ASUSTekCOMPU_ec:0e:...	Broadcast	ARP	42	Who has 192.168.10.1? Tell 192.168.10.4 (duplicate use of 192.168.10.4...)

3. Opening Detailed Information

Investigated the detailed information of the duplicate ARP packets to gather more context about the attack. This included examining the source and destination MAC addresses and IP addresses.

```
Frame 26: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{CCC4B960-1E92-4BD5-11D4-000000000000}
Ethernet II, Src: ASUSTekCOMPU_ec:0e:7f (50:eb:f6:ec:0e:7f), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Address Resolution Protocol (request)
[Duplicate IP address detected for 192.168.10.4 (50:eb:f6:ec:0e:7f) - also in use by 08:00:27:53:0c:ba (frame 25)]
  [Frame showing earlier use of IP address: 25]
    [Expert Info (Warning/Sequence): Duplicate IP address configured (192.168.10.4)]
      [Duplicate IP address configured (192.168.10.4)]
        [Severity level: Warning]
        [Group: Sequence]
    [Seconds since earlier frame seen: 1]
```

4. Identifying the Attacker's IP

Identified the real IP address of the attacker by correlating the ARP responses with legitimate network traffic. This involved checking for inconsistencies in MAC address mappings.

```
Frame 19: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{CCC4B960-1E92-4BD5-11D4-000000000000}
Ethernet II, Src: PCSSystemtec_53:0c:ba (08:00:27:53:0c:ba), Dst: Netgear_e2:d5:c3 (2c:30:33:e2:d5:c3)
Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: PCSSystemtec_53:0c:ba (08:00:27:53:0c:ba)
  Sender IP address: 192.168.10.4
  Target MAC address: Netgear_e2:d5:c3 (2c:30:33:e2:d5:c3)
  Target IP address: 192.168.10.1
```

5. Observing Traffic Drop

Noticed a drop in network traffic, indicating a lack of proper forwarding. This confirmed that the ARP spoofing was causing a denial-of-service (DoS) condition as the attacker was not forwarding the intercepted traffic correctly.

Time	Source	Destination	Protocol	Details
35.63.215761	PCSSystemtec_53:0c:ba	Netgear_e2:d5:c3	ARP	60 192.168.10.4 is at 08:00:27:53:0c:ba
36.64.016999	204.79.197.254	192.168.10.4	TCP	60 443 → 50985 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37.64.307719	52.212.57.111	192.168.10.4	TLSv1.2	85 Encrypted Alert
38.64.441145	52.212.57.111	192.168.10.4	TLSv1.2	85 Encrypted Alert
39.65.043801	52.212.57.111	192.168.10.4	TCP	85 [TCP Retransmission] 443 → 51811 [FIN, PSH, ACK] Seq=1 Ack=1
40.65.177343	52.212.57.111	192.168.10.4	TCP	85 [TCP Retransmission] 443 → 64208 [FIN, PSH, ACK] Seq=1 Ack=1
41.65.375455	PCSSystemtec_53:0c:ba	Netgear_e2:d5:c3	ARP	60 192.168.10.4 is at 50:eb:f6:ec:0e:7f
42.66.278704	52.51.147.233	192.168.10.4	TCP	60 443 → 51046 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1460

Mitigation

To mitigate ARP spoofing attacks and prevent similar incidents in the future, several measures can be implemented

Static ARP Entries

Configure static ARP entries on critical systems and network devices. This ensures that important IP-MAC address mappings are fixed and cannot be altered by malicious ARP replies.

ARP Spoofing Detection Tools

Utilize ARP spoofing detection tools and intrusion detection systems (IDS) to monitor and alert on suspicious ARP traffic. Tools like arpswatch or ArpON can be used to keep track of ARP changes in the network.

Port Security

Implement port security on switches to restrict the number of MAC addresses that can be learned on a single port. This can prevent an attacker from introducing new MAC addresses into the network.

Dynamic ARP Inspection (DAI)

Enable Dynamic ARP Inspection on switches. DAI validates ARP packets against trusted DHCP snooping binding tables, ensuring only legitimate ARP requests and responses are allowed.

VLAN Segmentation

Segment the network into smaller VLANs to limit the broadcast domain and reduce the impact of ARP spoofing attacks. This can also make it easier to detect and isolate suspicious activity.

Conclusion

Detecting ARP spoofing is crucial for maintaining network integrity and availability. This lab demonstrated how to use Wireshark to identify ARP spoofing by analyzing duplicate ARP entries, finding the attacker's real IP address, and recognizing the resulting traffic disruption. Implementing effective mitigation strategies, such as static ARP entries, ARP spoofing detection tools, port security, Dynamic ARP Inspection, VLAN segmentation, and regular audits, can significantly reduce the risk and impact of ARP spoofing attacks on network infrastructure.